Soft Projective Reassembly of the Physical Row Mask: Orthogonal GCD Compression, Optimal Ferrers Separation, and Coherent Product–Slope Spectra

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Abstract

The preceding orbit–sliced analysis leaves a concrete interface question: can the actual row-pair multiplier be reassembled into separable layers at only $X^{o(1)}$ cost, without destroying the product slope that carries the multiplicative spectrum? We answer the static part affirmatively and locate the next obstruction exactly.

For a finite kernel K, let $\pi_\infty(K)$ be its bounded projective norm. The physical row mask is

$$\mathfrak{m}_{\mathrm{act}}(\alpha,\gamma)=\mathbf{1}_{\ell_\alpha\neq\ell_\gamma}\mathbf{1}_{|m_\alpha-m_\gamma|>\Delta}\mathbf{1}_{(d_\alpha,d_\gamma)\leq G}.$$

The same-source deletion has projective cost at most 2. The near-row deletion is a finite interval kernel and has cost $O(\log(2Q))$. For the gcd threshold, an orthogonal random-sign compression of the exact divisor projector gives

$$\pi_\infty\big(\mathbf{1}_{(d,e)\leq G}\big)\leq(3+2\sqrt{2})^{\omega_*},\qquad \omega_*:=\max_{\text{physical }d}\omega(d).$$

Consequently

$$\boxed{\pi_\infty(\mathfrak{m}_{\mathrm{act}})\ll\log(2Q)(3+2\sqrt{2})^{\omega_*}=X^{o(1)}.}$$

Together with the previously proved finite-residue and Mellin separations, this closes the static controlled-multiplier boundary in the retained TPC-32 factorization. It is a projective-complexity theorem, not an operator-norm bound for the physical Gram matrix.

Fixing the opposite row gives a stronger product-ready form. The gcd and same-source conditions become unary weights, while the near-row cutoff is a difference of two Ferrers kernels. After splitting the $O_h(1)$ fixed orbit residue cells, finite Fourier inversion therefore decomposes each physical column exactly into

$$b_\rho(\ell)f_\rho(d)\eta_\rho(j)$$

with $O(D)$ discrete layers but total signed projective and orbit-seminorm cost $X^{o(1)}$; in particular b_ρ is independent of j and of the complementary factor s. The logarithmic cost is optimal: for $T_n=(\mathbf{1}_{k\leq r})$,

$$\pi_\infty(T_n)\asymp\log(2n),\qquad \pi_+(T_n)=n.$$

Thus signed Fourier reassembly is essential, whereas positive or raw column-by-column rectangle splitting is polynomially expensive.

Termwise additive-frequency separation of the product phase is costly. For $A_r(x,y)=e_q(rxy)$ on $\mathbb{F}_q^\times$, $r\neq 0$,

$$\boxed{\pi_\infty(A_r)=\frac{(q-2)\sqrt{q}+1}{q-1}\asymp\sqrt{q}.}$$

By contrast the multiplication kernel $\mathbf{1}_{xy=a}$ has projective cost exactly 1. Expanding that kernel additively and reassembling its frequencies absolutely manufactures a $\sqrt{q}$ amplitude loss, which at $q \asymp J$ is precisely one factor J in energy.

Finally, we diagonalize the coherent complementary-factor sum. For the TPC-35 product–slope operator T and an arbitrary weight w,

$$\sum_j \left| \sum_s w(s)(Tb)(j,s) \right|^2 = \frac{1}{q-1} \sum_\chi |\widehat{b}(\bar{\chi})|^2 \left| \sum_s w(s) \widehat{R}_s(\chi) \right|^2 .$$

If f, g are additively centered and $w = 1$, then

$$\sum_s R_s(a) = qg(0)f(-h/a),$$

so complete s-coherence collapses to a residue-zero boundary trace. Nevertheless the factor $q-1$ from Cauchy in s is sharp in the abstract centered nonprincipal class. The remaining physical gate is therefore not static-mask separation: it is the joint control of incomplete s-weights, integer aliases, and the actual Möbius coefficient mass in the resulting character spectrum. We prove no coefficient-specific orbit gate, affine Chowla theorem, parity breach, Hardy–Littlewood asymptotic, or twin–prime result.

Keywords: Möbius correlation; projective tensor norm; Ferrers matrix; multiplicative character; orbit energy; parity boundary.

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1 Introduction

The matched residual chain has reached a useful separation of tasks. TPC-32 keeps the exact rank–two cutoff polarization but leaves the joint row multiplier inside a bilinear form. TPC-33 opens one ultra divisor and obtains a fixed-determinant affine column after the opposite row is fixed. TPC-34 converts the distinguished shell to orbit–sliced row energies, and TPC-35 identifies the coefficient-specific multiplicative spectrum that would be available after complete product–slope reassembly [4–7].

The inherited sufficient gate is

$$V_L + V_R \ll X^{o(1)} \frac{Q^3}{J}, \tag{1}$$

whereas the proved Gram-copy diagonal is

$$V_{L,\Delta} + V_{R,\Delta} \ll X^{o(1)} Q^2 J. \tag{2}$$

At the endpoint

$$Q = X^{267/400+o(1)}, \qquad J = X^{133/400+o(1)}, \qquad T = X^{193/500+o(1)}, \tag{3}$$

the allowance between (1) and (2) is only

$$\frac{Q}{J^2} = X^{1/400+o(1)}. \tag{4}$$

One cannot afford a new fixed power of X.

TPC-35 therefore left a precise preliminary question: does the actual physical multiplier admit a signed separable representation of soft cost, or is the multiplier itself another polynomial obstruction? The answer is positive. The more delicate conclusion is that this does not close (1); it relocates the gap to completion, aliases, and coherent complementary-factor spectra.

1.1 The static multiplier is softly projective

For a kernel K on two finite sets, write $\pi_\infty(K)$ for its bounded projective norm; see section 3. The physical static mask has three factors:

$$\mathfrak{m}_{\mathrm{act}}(\alpha,\gamma) = \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > \Delta} \mathbf{1}_{(d_\alpha, d_\gamma) \leq G}, \qquad \Delta = QX^{-\kappa_0}, \quad G = X^{\kappa_0}. \tag{5}$$

The same-source factor is compressed by source orthogonality. The near-row factor is a one-dimensional interval kernel. The new step is a global orthogonal compression of the row-gcd projector. If $\omega_* = \max_\alpha \omega(d_\alpha)$, then

$$\boxed{\pi_\infty(\mathfrak{m}_{\mathrm{act}}) \ll \log(2Q)(3+2\sqrt{2})^{\omega_*} = X^{o(1)}.} \tag{6}$$

The equality $X^{o(1)}$ uses only the standard fixed-base divisor estimate $C^{\omega(n)} \ll_{C,\varepsilon} n^\varepsilon$. It is uniform in the pruning threshold G.

The distinction between layer count and projective mass is essential. The exact finite representation may contain polynomially many layers, but their normalized total variation is soft. Every estimate uniform in the bounded factors may therefore be integrated without paying for the raw number of layers.

1.2 Columnwise separation and its optimal cost

The orbit energy first fixes the opposite row. For $\gamma = (\ell_\gamma, e)$, the gcd cutoff in (5) is then a unary d-weight. Ordering the physical d-values turns the near-row condition into the difference of two prefix, or Ferrers, kernels. After splitting the fixed $O_h(1)$ orbit residue cells, finite Fourier inversion gives the exact representation

$$\gamma^{(1)}_{(\ell,d)} \mathfrak{A}^{\mathrm{act}}_{(\ell,d),\gamma}(j) = \sum_{r \bmod H_0} \mathbf{1}_{j \equiv r \pmod{H_0}} \int \sum_\rho c_{\rho,\omega,\gamma,r} b_{\rho,\omega,\gamma,r}(\ell) f_{\rho,\omega,\gamma,r}(d) \eta_{\rho,\omega,\gamma,r}(j/J)\, d\nu_r(\omega), \tag{7}$$

with $O(D)$ discrete layers and $X^{o(1)}$ total signed orbit-seminorm cost. In particular $b(\ell)$ is independent of j and of the reflected complementary factor s. This proves the static product-ready shape asked for in TPC-35.

The logarithm is not an artifact. For the triangular Ferrers matrix

$$T_n = (\mathbf{1}_{k \leq r})_{1 \leq r,k \leq n}, \tag{8}$$

we prove

$$\pi_\infty(T_n) \asymp \log(2n). \tag{9}$$

If every rank-one layer is required to be nonnegative, its exact cost is instead n. Thus requiring every rectangle layer to be entrywise nonnegative destroys the gain that signed Fourier orthogonality supplies.

1.3 The cost of absolute additive-frequency separation

The product slope is not another harmless smooth factor. For prime q and $r \neq 0$, put

$$A_r(x,y) = e_q(rxy), \qquad x,y \in \mathbb{F}_q^\times. \tag{10}$$

Multiplicative characters give a flat singular-vector basis and the exact projective norm

$$\boxed{\pi_\infty(A_r) = \frac{(q-2)\sqrt{q}+1}{q-1}.} \tag{11}$$

On the other hand, $K_a(x, y) = \mathbf{1}_{xy=a}$ is a permutation kernel and $\pi_\infty(K_a) = 1$. The additive delta expansion of K_a, followed by termwise absolute reassembly of its nonzero frequencies, costs $\asymp \sqrt{q}$. At $q \asymp J$, this is exactly a factor J after squaring. Thus the additive-frequency expansion cannot be estimated at the critical scale by summing its frequency costs separately. The multiplicative-character reconstruction keeps projective cost one.

1.4 The coherent complementary-factor spectrum

TPC-35 diagonalizes the complete square energy $\sum_{j,s} |(Tb)(j, s)|^2$. The physical column first sums over the complementary factor s and then squares. We prove the corresponding weighted identity. For $G_q = \mathbb{F}_q^\times$, any $w : G_q \to \mathbb{C}$, and

$$S_w(j) = \sum_{s \in G_q} w(s)(Tb)(j, s), \tag{12}$$

one has

$$\boxed{\sum_j |S_w(j)|^2 = \frac{1}{q-1} \sum_\chi |\widehat{b}(\bar{\chi})|^2 \left| \sum_s w(s) \widehat{R}_s(\chi) \right|^2.} \tag{13}$$

For the complete weight $w = 1$, additive centering gives the exact trace

$$\sum_{s \in G_q} R_s(a) = qg(0)f(-h/a). \tag{14}$$

Thus full s-coherence collapses to a residue-zero boundary term. Yet there are centered nonprincipal data for which $(Tb)(j, s)$ is independent of s, so the factor $q - 1$ from Cauchy in s is sharp. A physical gain must use the actual Möbius-derived g, the actual incomplete weight w, or joint alias cancellation.

1.5 Claim boundary and organization

The paper resolves a missing interface, not the twin-prime problem. It proves soft projective complexity of the actual static mask, an exact columnwise product-ready decomposition, two sharp order-of-operations obstructions, and coherent finite-field spectra. It does not prove a small physical Gram norm, the coefficient gate (1), a three-modulus alias theorem, an affine Chowla estimate, positivity, or a Hardy–Littlewood asymptotic.

Section 2 fixes the inherited object. Section 3 develops the projective tools. Section 4 proves (6). Section 5 proves the product-ready column theorem. Section 6 gives the sharp Ferrers and product-phase boundaries. Section 7 proves (13)– (14). The last two sections record the endpoint ledger, next gate, exact certificate, and formal claim boundary.

2 The inherited physical interface

We import the actual packet and proved orbit reduction from TPC-25 and TPC-32–TPC-35 [3–7]. This section makes clear which object is reassembled and which later completion remains open.

2.1 Rows, coefficients, and the joint multiplier

Fix $h \in \mathbb{Z} \setminus \{0\}$, put $H = \operatorname{rad}|h|$, and work on one source-compatible dyadic packet with

$$Q = LD, \qquad QJ \asymp X, \qquad R \le T < U_0 \asymp X. \tag{15}$$

A row is

$$\alpha = (\ell_\alpha, d_\alpha), \qquad m_\alpha = \ell_\alpha d_\alpha \asymp Q, \tag{16}$$

where $\ell_\alpha \asymp L$ is prime, $d_\alpha \asymp D$ is squarefree, and $(d_\alpha, H) = 1$. The source separation

$$R = X^{1/2-\delta+o(1)}, \qquad D \le R^{1/2+o(1)}, \qquad L > \max\{2R, 2|h|\} \tag{17}$$

makes the integer m_α determine its row label uniquely.

The two physical row coefficients are

$$\gamma_\alpha^{(i)} = \mu(d_\alpha)(\log \ell_\alpha)\omega_D^{(i)}(d_\alpha)\psi_L^{(i)}(\ell_\alpha/L)\zeta_\alpha^{(i)}, \qquad i = 1, 2, \tag{18}$$

with fixed smooth and finite-residue factors. The inherited masses are

$$\sum_\alpha |\gamma_\alpha^{(i)}| \ll X^{o(1)}Q, \tag{19}$$

$$\sum_\alpha |\gamma_\alpha^{(i)}|^2 \ll X^{o(1)}Q. \tag{20}$$

The logarithm in the original L^2 estimate is absorbed into $X^{o(1)}$.

For j in an interval of $O(J)$ integers, put

$$N_\alpha(j) = m_\alpha j + h \asymp X. \tag{21}$$

On the physical support every divisor $u \mid N_\alpha(j)$ satisfies

$$(u, m_\alpha j H) = 1. \tag{22}$$

Let

$$\Delta = QX^{-\kappa_0}, \qquad G = X^{\kappa_0}, \tag{23}$$

where $\kappa_0 > 0$ is the inherited pruning parameter. The actual static mask is

$$\boxed{\mathfrak{m}_{\mathrm{act}}(\alpha, \gamma) = \mathbf{1}_{\ell_\alpha \ne \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > \Delta} \mathbf{1}_{(d_\alpha, d_\gamma) \le G}.} \tag{24}$$

The full multiplier is

$$\mathfrak{A}^{\mathrm{act}}_{\alpha,\gamma}(j) = \mathfrak{m}_{\mathrm{act}}(\alpha, \gamma)\Xi_{\alpha,\gamma}(j)\mathcal{W}_{\alpha,\gamma}(j/J). \tag{25}$$

Let $H_0 = O_h(1)$ be a common period in j for the fixed-residue factor Ξ. TPC-25 proves that Ξ may be handled by first splitting the $O_h(1)$ cells $j \bmod H_0$, and that on each cell the smooth factor $\mathcal{W}$ has an absolutely convergent Mellin projective separation with polynomially weighted orbit-seminorm mass bounded independently of X. The new issue is therefore the nonsmooth mask (24).

2.2 Orbit slicing and reflected columns

Let

$$C_m(j) = \sum_{\substack{T < u \le U_0 \\ u \mid mj+h}} -\mu(u) \log u. \tag{26}$$

The left orbit slice is

$$Y^L_{\gamma,j} = \sum_\alpha \gamma_\alpha^{(1)} \mathfrak{A}^{\mathrm{act}}_{\alpha,\gamma}(j) C_{m_\alpha}(j), \qquad V_L = \sum_{\gamma,j} |Y^L_{\gamma,j}|^2, \tag{27}$$

and the right slice exchanges the two row systems. TPC-34 proves that the bound (1) is sufficient for the remaining scalar shell and proves the diagonal (2).

Open the ultra divisor in (27). For a nonzero term write

$$\ell dj + h = su, \qquad T < u \le U_0, \qquad 1 \le s \ll X/T. \tag{28}$$

After one fixes γ, a product-ready row decomposition would turn the inner contribution into layers of the schematic form

$$\mathcal{Z}_{\gamma,\rho}(j,s) = \sum_{\ell,d,u} b_{\gamma,\rho}(\ell) f_{\gamma,\rho}(d) g_{\gamma,\rho}(u) \mathbf{1}_{su=\ell jd+h}. \tag{29}$$

The goal of section 5 is to prove the exact separation in ℓ, d, j that precedes (29).

2.3 What static reassembly does not include

Modulo a prime $q \nmid h$, a complete version of (29) becomes

$$R_s(a) = \sum_{D\in\mathbb{F}_q} f(D) g\Big(s^{-1}(aD+h)\Big), \tag{30}$$

$$(Tb)(j,s) = \sum_{\ell\in\mathbb{F}_q^\times} b(\ell) R_s(\ell j). \tag{31}$$

TPC-35 gives the exact character spectrum of (31). Passing from (29) to that complete finite-field operator additionally requires:

(i) controlling aliases between the integer equation and its modular relaxation;

(ii) completing the physical j, d, u, s ranges without a polynomial cost; and

(iii) retaining the coherent s-sum in the topology of (27).

None of these statements follows merely from a low projective norm for (24). We keep them separate throughout.

2.4 Endpoint scales

At the high–β endpoint,

$$L = X^{99979/210000+o(1)}, \qquad D = X^{10049/52500+o(1)}, \tag{32}$$

in addition to (3). Hence

$$D = o(J), \qquad J = o(Q), \qquad QJ = X^{1+o(1)}. \tag{33}$$

The number of formal column layers $O(D)$ below is polynomial, but its exponent is irrelevant to estimates that reassemble by signed projective mass. Counting layers absolutely would be the wrong norm.

3 Bounded projective norms and finite Fourier compression

The relevant complexity is the normalized mass of a signed rank-one representation, not its number of summands.

3.1 The bounded projective norm

Definition 3.1 (Bounded projective norm)**.** For finite sets $\mathcal{X}, \mathcal{Y}$ and a kernel $K : \mathcal{X} \times \mathcal{Y} \to \mathbb{C}$, define

$$\pi_\infty(K) := \inf \int_\Omega \|p_\omega\|_{\ell^\infty(\mathcal{X})} \|q_\omega\|_{\ell^\infty(\mathcal{Y})} \, d|\nu|(\omega), \tag{34}$$

where the infimum runs over all finite signed or complex measures and representations

$$K(x, y) = \int_\Omega p_\omega(x) q_\omega(y) \, d\nu(\omega). \tag{35}$$

Finite sums suffice in finite dimensions; probability-space notation is convenient for orthogonal compression.

The normalization permits the scalar coefficient of a layer to be placed in either factor or in the measure. The following properties will be used repeatedly.

Lemma 3.2 (Elementary projective calculus)**.** *For finite kernels K, L on the same product set,*

$$\|K\|_\infty \le \pi_\infty(K), \tag{36}$$

$$\pi_\infty(K + L) \le \pi_\infty(K) + \pi_\infty(L), \tag{37}$$

$$\pi_\infty(K \circ L) \le \pi_\infty(K)\pi_\infty(L), \tag{38}$$

where $\circ$ is entrywise multiplication. If K is viewed as an $m \times n$ matrix, then

$$\boxed{\pi_\infty(K) \ge \frac{\|K\|_*}{\sqrt{mn}},} \tag{39}$$

where $\|K\|_*$ is its nuclear norm.

Proof. The first two assertions follow directly from (35). Multiplying two such representations over the product measure proves (38). A rank-one matrix pq^T has nuclear norm $\|p\|_2\|q\|_2$, which is at most $\sqrt{mn}\|p\|_\infty\|q\|_\infty$. Integrate and take the infimum. □

Remark 3.3 (Projective complexity is not a Gram estimate)*.* The norm π_∞ controls the cost of passing a bilinear estimate through a Schur multiplier. It does not imply that the multiplier matrix, or the off-diagonal Gram matrix obtained after inserting it, has small operator norm. In particular, (39) is used below only for finite-model optimality statements.

3.2 A discrete interval lemma

Write $e_M(t) = \exp(2\pi i t/M)$. For $1 \le N < M/2$, define

$$\kappa_N(t) = \mathbf{1}_{\{0,1,\dots,N-1\}}(t), \qquad t \in \mathbb{Z}/M\mathbb{Z}, \tag{40}$$

and use the normalized Fourier transform

$$\widehat{\kappa}_N(a) = \frac{1}{M} \sum_{t=0}^{N-1} e_M(-at). \tag{41}$$

Lemma 3.4 (Discrete interval Fourier mass)**.** *Uniformly for $1 \le N < M/2$,*

$$\sum_{a \bmod M} |\widehat{\kappa}_N(a)| \ll 1 + \log(2N). \tag{42}$$

Consequently the prefix kernel

$$H_N(k, r) = \mathbf{1}_{1 \le r \le k}, \qquad 0 \le k \le N, \quad 1 \le r \le N, \tag{43}$$

has the exact representation

$$H_N(k,r) = \sum_{a \bmod M} \widehat{\kappa}_N(a) e_M(ak) e_M(-ar) \tag{44}$$

and satisfies

$$\pi_\infty(H_N) \ll 1 + \log(2N). \tag{45}$$

Proof. For $a \neq 0$, the geometric sum gives

$$|\widehat{\kappa}_N(a)| \leq \min\left\{\frac{N}{M}, \frac{1}{2\min(a, M-a)}\right\}, \tag{46}$$

after taking $1 \leq a \leq M-1$. The terms with $\min(a, M-a) \leq M/N$ contribute $O(1)$; the remaining harmonic sum is $O(1+\log N)$. Fourier inversion proves (44), because $r \leq k$ is equivalent to $k - r \in \{0, \dots, N-1\}$ when $M \geq 2N+1$. The factors in that representation have modulus one, proving (45). □

3.3 Finite orthogonality as compression

If a finite set $\mathcal{I}$ is injected into $\mathbb{Z}/M\mathbb{Z}$, then

$$\frac{1}{M} \sum_{t \bmod M} e_M\big(t(\iota(v) - \iota(w))\big) = \mathbf{1}_{v=w}. \tag{47}$$

This identity will compress a sum over common divisors. The number M of formal phases can be large; the normalized measure $M^{-1}\sum_t$ has mass one. Replacing (47) by independent Rademacher signs gives the same projective bound.

4 Global soft reassembly of the actual static mask

We now handle the three factors of (24). Every identity in this section is pointwise on the finite physical row set.

4.1 Same-source deletion

Let $\mathcal{L}$ be the finite set of prime sources in the packet. Inject it into $\mathbb{Z}/M_L\mathbb{Z}$. Orthogonality gives

$$\mathbf{1}_{\ell=\ell'} = \frac{1}{M_L} \sum_{t \bmod M_L} e_{M_L}(t\iota(\ell)) e_{M_L}(-t\iota(\ell')). \tag{48}$$

Hence

$$\pi_\infty(\mathbf{1}_{\ell=\ell'}) = 1, \qquad \pi_\infty(\mathbf{1}_{\ell\neq\ell'}) \leq 2. \tag{49}$$

The equality in the first formula follows also from the entrywise lower bound. There is no factor equal to the number of prime sources.

4.2 Near-row deletion

Order the N physical rows so that

$$m_1 < m_2 < \cdots < m_N, \tag{50}$$

which is possible by the inherited row-label injectivity. Write $r(\alpha)$ for the rank of m_α. For a fixed right row γ, define

$$k_-(\gamma) := \#\{\alpha : m_\alpha < m_\gamma - \Delta\}, \tag{51}$$

$$k_+(\gamma) := \#\{\alpha : m_\alpha \le m_\gamma + \Delta\}. \tag{52}$$

Then, including all endpoint conventions exactly,

$$\boxed{\mathbf{1}_{|m_\alpha - m_\gamma| > \Delta} = 1 + H_N(k_-(\gamma), r(\alpha)) - H_N(k_+(\gamma), r(\alpha)).} \tag{53}$$

Proposition 4.1 (Near-row Fourier reassembly)**.** *For every* $\Delta \ge 0$*,*

$$\pi_\infty\Big(\mathbf{1}_{|m_\alpha - m_\gamma| > \Delta}\Big) \ll 1 + \log(2N) \ll \log(2Q). \tag{54}$$

Proof. Insert (44) into the two prefix kernels in (53), then apply the triangle inequality. The physical row count is $N \ll QX^{o(1)}$, so $\log(2N) \ll \log(2Q)$. □

The proof uses only the order of the row labels. It does not require the near band to have constant width or the labels to fill an integer interval.

4.3 Orthogonal compression of the row-gcd projector

TPC-33 proves the exact identity

$$\mathbf{1}_{(d,e)\le G} = \sum_{\substack{v|d\\ v|e}} \lambda_G(v), \qquad \lambda_G(v) = \sum_{\substack{g|v\\ g\le G}} \mu(v/g). \tag{55}$$

We upgrade its fixed-column absolute complexity to a global signed projective representation.

Let $\mathcal{I}_G$ be the finite set of v dividing at least one physical squarefree row coordinate, and inject $\mathcal{I}_G$ into $\mathbb{Z}/M_G\mathbb{Z}$. Put

$$P_t(d) := \sum_{v|d} \operatorname{sgn}(\lambda_G(v))\sqrt{|\lambda_G(v)|}\, e_{M_G}(t\iota(v)), \tag{56}$$

$$Q_t(e) := \sum_{v|e} \sqrt{|\lambda_G(v)|}\, e_{M_G}(-t\iota(v)), \tag{57}$$

with $\operatorname{sgn}(0) = 0$.

Theorem 4.2 (Global orthogonal gcd compression)**.** *For physical squarefree* d, e*,*

$$\boxed{\mathbf{1}_{(d,e)\le G} = \frac{1}{M_G} \sum_{t \bmod M_G} P_t(d)Q_t(e).} \tag{58}$$

If $\omega_* = \max\{\omega(d) : d$ *occurs in the packet*$\}$*, then*

$$\boxed{\pi_\infty\Big(\mathbf{1}_{(d,e)\le G}\Big) \le (3 + 2\sqrt{2})^{\omega_*} = X^{o(1)}.} \tag{59}$$

Proof. Insert (56)–(57), average over t, and apply (47). Only equal divisors survive, giving $\sum_{v|d,e} \lambda_G(v)$, which is (55).

For squarefree v,

$$|\lambda_G(v)| \le \sum_{g|v} |\mu(v/g)| = 2^{\omega(v)}. \tag{60}$$

Therefore, uniformly in t,

$$|P_t(d)| \le \sum_{v|d} 2^{\omega(v)/2} = (1 + \sqrt{2})^{\omega(d)}, \tag{61}$$

$$|Q_t(e)| \le (1 + \sqrt{2})^{\omega(e)}. \tag{62}$$

The normalized counting measure in (58) has total mass one. Multiplying the two suprema gives $(1 + \sqrt{2})^{2\omega_*} = (3 + 2\sqrt{2})^{\omega_*}$. Finally, $C^{\omega(n)} \ll_{C,\varepsilon} n^\varepsilon$ for each fixed $C > 1$, and $d, e \ll X^{O(1)}$. □

Remark 4.3 (Why absolute divisor expansion looked expensive). The raw set $\mathcal{I}_G$ can have polynomial size. Summing one rank-one term per v in absolute value ignores orthogonality between divisor coordinates. The average in (58) has the same number of formal phases but normalized mass one; only the divisor suprema remain.

4.4 The complete static-mask theorem

Theorem 4.4 (Soft projective reassembly of the actual mask). *On every inherited physical packet,*

$$\boxed{\pi_\infty(\mathfrak{m}_{\mathrm{act}}) \ll \log(2Q)(3+2\sqrt{2})^{\omega_*} = X^{o(1)}.} \tag{63}$$

The estimate is uniform in the fixed pruning parameter and its threshold $G = X^{\kappa_0}$.

Proof. The mask (24) is the Schur product of the three kernels treated in (49), (54), and (59). Apply (38). □

Corollary 4.5 (The retained-multiplier interface). *After multiplication by the actual fixed-period and smooth factors in (25), the joint multiplier has the residue-cell representation*

$$\mathfrak{A}^{\mathrm{act}}_{\alpha,\gamma}(j) = \sum_{r \bmod H_0} \mathbf{1}_{j\equiv r \pmod{H_0}} \int_{\Omega_r} p_{\omega,r}(\alpha) q_{\omega,r}(\gamma) \phi_{\omega,r}(j/J)\, d\nu_r(\omega) \tag{64}$$

such that, for every fixed $B \geq 0$,

$$\sum_{r \bmod H_0} \int_{\Omega_r} \|p_{\omega,r}\|_\infty \|q_{\omega,r}\|_\infty \|\phi_{\omega,r}\|_{C^B}\, d|\nu_r|(\omega) \ll_B X^{o(1)}. \tag{65}$$

Thus, after the fixed $O_h(1)$ residue-cell split, the conditional projective product representation of TPC-32 is unconditional for the actual multiplier on this packet.

Proof. TPC-25 supplies the finite residue-cell split and, on each cell, an absolutely convergent Mellin separation for the smooth row–orbit factors, with every fixed orbit seminorm integrable [3]. The periodic j-factor is constant on a cell and is not differentiated as a function of j/J. Take the product with the representation in theorem 4.4; the static factors do not change the orbit seminorms. Summing the fixed number of cells proves the claim. □

Corollary 4.6 (A trace-norm consequence). *If the physical row set has cardinality N, then*

$$\|\mathfrak{m}_{\mathrm{act}}\|_* \leq N\pi_\infty(\mathfrak{m}_{\mathrm{act}}) \ll QX^{o(1)}. \tag{66}$$

This is an upper bound for the static mask matrix. It is not an upper bound of size Q^2/J for either orbit-Gram off-diagonal matrix, whose entries contain an additional sum over time and arithmetic coefficients.

5 Fixed-column product-ready reassembly

The global theorem separates the two row labels, but a global factor may still depend jointly on the source ℓ and divisor coordinate d through $m = \ell d$. The product–slope operator needs one further separation. Orbit slicing provides the correct order: fix the opposite row first.

5.1 The exact Ferrers formula

Fix

$$\gamma = (\ell_\gamma, e), \qquad n = m_\gamma = \ell_\gamma e. \tag{67}$$

Let $\mathcal{D} = \{d_1 < \cdots < d_{N_D}\}$ be an ambient integer set containing the physical d-support. One may take the full dyadic integer interval, so

$$N_D \ll D. \tag{68}$$

Squarefreeness, coprimality, and smooth support are retained later as unary d-weights. Write $r(d_k) = k$. For every source ℓ, put

$$k_-^\gamma(\ell) := \#\{d \in \mathcal{D} : \ell d < n - \Delta\}, \tag{69}$$

$$k_+^\gamma(\ell) := \#\{d \in \mathcal{D} : \ell d \le n + \Delta\}. \tag{70}$$

Then

$$\mathbf{1}_{|\ell d - n| > \Delta} = 1 + H_{N_D}(k_-^\gamma(\ell), r(d)) - H_{N_D}(k_+^\gamma(\ell), r(d)). \tag{71}$$

Choose $M_D = 2N_D + 1$, and let $\widehat{\kappa}_{N_D}$ be as in (41). Combining (71) with (44) gives the pointwise identity

$$\begin{aligned} &\mathfrak{m}_{\mathrm{act}}((\ell, d), \gamma) \\ &= \mathbf{1}_{\ell \ne \ell_\gamma} \mathbf{1}_{(d,e) \le G} \left[1 + \sum_{a \bmod M_D} \widehat{\kappa}_{N_D}(a) (e_{M_D}(a k_-^\gamma(\ell)) - e_{M_D}(a k_+^\gamma(\ell))) e_{M_D}(-a r(d)) \right]. \end{aligned} \tag{72}$$

Every factor outside the brackets is unary once γ is fixed.

Theorem 5.1 (Exact fixed-column source–divisor separation)**.** *For every fixed opposite row γ, there are coefficients and bounded functions such that*

$$\mathfrak{m}_{\mathrm{act}}((\ell, d), \gamma) = \sum_{\rho \in \mathcal{R}_\gamma} c_{\rho,\gamma} b_{\rho,\gamma}(\ell) f_{\rho,\gamma}(d), \tag{73}$$

with

$$|\mathcal{R}_\gamma| \ll D, \tag{74}$$

$$\sum_{\rho \in \mathcal{R}_\gamma} |c_{\rho,\gamma}| \|b_{\rho,\gamma}\|_\infty \|f_{\rho,\gamma}\|_\infty \ll 1 + \log(2D) = X^{o(1)}. \tag{75}$$

The estimate is uniform in γ.

Proof. Use the constant term and the M_D Fourier terms displayed in (72). The source factor and the two threshold phases belong to b; the gcd cutoff and rank phase belong to f. Their sup norms are bounded by 2 and 1, respectively. The layer count is $1 + M_D = O(D)$, while (42) proves the signed mass bound. Importantly, the normalized Fourier coefficients are summed in absolute value; the raw layer count is not. □

5.2 Including the physical row and orbit weights

The coefficient (18) is already a product of a source weight and a d-weight, apart from a fixed-period factor. First split the $O_h(1)$ orbit cells $j \bmod H_0$; finite Fourier inversion in the remaining row residue variables has bounded total mass. On each cell, Mellin inversion separates every smooth factor $W(\ell d j / X)$ into $\ell^{it} d^{it} j^{it}$ and has Schwartz total variation [3]. Multiplying these inherited representations by (73), and enlarging $\mathcal{R}_\gamma$ by only an $O_h(1)$ factor, proves the physical form below.

Theorem 5.2 (Physical product-ready column)**.** *For each fixed γ, the actual left-column multiplier including its left row coefficient has the exact residue-cell representation*

$$\gamma^{(1)}_{(\ell,d)}\mathfrak{A}^{\mathrm{act}}_{(\ell,d),\gamma}(j) = \sum_{r \bmod H_0} \mathbf{1}_{j\equiv r \pmod{H_0}} \int_{\Omega_r} \sum_{\rho\in\mathcal{R}_\gamma} c_{\rho,\omega,\gamma,r} b_{\rho,\omega,\gamma,r}(\ell) f_{\rho,\omega,\gamma,r}(d) \eta_{\rho,\omega,\gamma,r}(j/J)\, d\nu_r(\omega). \tag{76}$$

For every fixed $B \geq 0$, its normalized mass satisfies

$$\sum_{r \bmod H_0} \int_{\Omega_r} \sum_{\rho\in\mathcal{R}_\gamma} |c_{\rho,\omega,\gamma,r}| \|b_{\rho,\omega,\gamma,r}\|_\infty \|f_{\rho,\omega,\gamma,r}\|_\infty \|\eta_{\rho,\omega,\gamma,r}\|_{C^B}\, d|\nu_r|(\omega) \ll_B X^{o(1)}, \tag{77}$$

after the fixed logarithmic source weight is included in the soft factor. For fixed r, the source function $b_{\rho,\omega,\gamma,r}(\ell)$ is independent of j, d, and the complementary factor s.

The statement is symmetric: fixing α instead gives the analogous right-column representation with the two row systems exchanged.

Remark 5.3 (Coefficient normalization). If the factor $\log \ell$ is not absorbed into $X^{o(1)}$, the right side of (77) acquires the explicit harmless factor $O(\log L)$. The theorem does not sum the physical row coefficients in ℓ^1; it controls the multiplier representation uniformly before the row sum.

5.3 Opening the ultra divisor

Insert (76) into (27), work on one of the fixed residue cells, and open (26). On each layer, all unary support and smooth factors may be absorbed into b, f, g, and (28) gives

$$\mathcal{Z}_{\gamma,\rho,\omega}(j,s) = \sum_{\ell,d,u} b_{\gamma,\rho,\omega}(\ell) f_{\gamma,\rho,\omega}(d) g_{\gamma,\rho,\omega}(u) \mathbf{1}_{su=\ell jd+h}. \tag{78}$$

This is the integer product-slope normal form. Reducing (78) modulo q produces (30)–(31); the source weight remains independent of j, s.

Corollary 5.4 (The static half of the TPC-35 reassembly gate)**.** *The actual fixed-column mask, row coefficient, fixed-period factors, and smooth orbit factors admit an exact $X^{o(1)}$-cost decomposition into product-slope-ready layers. What remains before the TPC-35 complete finite-field spectrum can be applied to (27) is modular completion and coherent alias reassembly, not static row-mask separation.*

5.4 Compatibility with the projected affine column

TPC-33 alternatively opens the gcd projector after fixing $\gamma = (\ell_\gamma, e)$. For $v \mid e$, write $d = vD'$. Applying theorem 5.1 on each projected D'-range gives at most

$$\sum_{v\mid e} O\left(1 + \frac{D}{v}\right) \ll DX^{o(1)} \tag{79}$$

formal layers. TPC-33 proves

$$\sum_{v\mid e} |\lambda_G(v)| \leq 3^{\omega(e)}, \tag{80}$$

so their total signed Ferrers mass is

$$\ll 3^{\omega(e)} \log(2D) = X^{o(1)}. \tag{81}$$

This version retains the affine determinant form of TPC-33, with product slope ℓjv. Since $v \ll D = o(J)$, multiplication by v is a permutation modulo every prime $q \asymp J$. The direct version above is shorter; the projected version confirms compatibility with the previous Möbius sign-fusion normal form.

6 Sharp reassembly and order-of-operations boundaries

The preceding constructions use cancellation between signed layers. We now show that their logarithmic behavior is sharp and that the product phase belongs to a different complexity class.

6.1 The Ferrers logarithm is optimal

Let

$$T_n = (\mathbf{1}_{k \le r})_{1 \le r, k \le n} \tag{82}$$

be the lower-triangular summation matrix.

Theorem 6.1 (Optimal signed Ferrers cost)**.** *There are absolute constants $c, C > 0$ such that*

$$c \log(2n) \le \pi_\infty(T_n) \le C \log(2n). \tag{83}$$

More precisely, the singular values of T_n are

$$\boxed{\sigma_k(T_n) = \left[2 \sin \frac{(2k-1)\pi}{4n+2}\right]^{-1}, \qquad 1 \le k \le n.} \tag{84}$$

Proof. The case $n = 1$ is immediate. Assume $n \ge 2$. The upper bound is (45). For the lower bound, the inverse of T_n is the lower bidiagonal first-difference matrix. Direct substitution of the discrete sine vectors diagonalizes $(T_n^{-1})^* T_n^{-1}$ and gives eigenvalues

$$4 \sin^2 \frac{(2k-1)\pi}{4n+2}, \qquad 1 \le k \le n, \tag{85}$$

which proves (84). Since $\sin x \le x$, the first $\lfloor n/2 \rfloor$ singular values give

$$\|T_n\|_* \gg n \sum_{k \le n/2} \frac{1}{2k-1} \gg n \log(2n). \tag{86}$$

Apply (39) with $m = n$. ☐

Thus the $O(\log D)$ cost in theorem 5.1 cannot be replaced uniformly by a constant for arbitrary threshold profiles.

6.2 Positive layers have linear cost

Define the nonnegative projective cost

$$\pi_+(K) := \inf \sum_\rho c_\rho, \tag{87}$$

where the infimum is over exact decompositions

$$K = \sum_\rho c_\rho x_\rho y_\rho^\mathsf{T}, \qquad c_\rho \ge 0, \quad 0 \le x_\rho, y_\rho \le 1 \tag{88}$$

entrywise.

Proposition 6.2 (Exact nonnegative Ferrers cost)**.** *For every $n \ge 1$,*

$$\boxed{\pi_+(T_n) = n.} \tag{89}$$

Proof. An exact nonnegative rank-one layer supported in the lower triangle can meet at most one diagonal entry. Indeed, if it is positive at both (i,i) and (j,j) with $i<j$, then its (i,j) entry is also positive, contradicting $(T_n)_{i,j}=0$. Since every diagonal entry of T_n equals one and the factors in (88) are at most one, summing the diagonal identities gives $\sum_\rho c_\rho \ge n$. Conversely,

$$T_n = \sum_{k=1}^{n} \mathbf{1}_{\{r:r\ge k\}} e_k^\mathsf{T} \tag{90}$$

has cost n. $\square$

This is the exact penalty for insisting that every intermediate layer be entrywise nonnegative. A raw column splitting or nonnegative rectangle reassembly can therefore lose D, whereas the signed proof loses only $X^{o(1)}$.

6.3 The product phase has square-root projective cost

Let $q>2$ be prime, put $G_q=\mathbb{F}_q^\times$, and for $r\in\mathbb{F}_q^\times$ define the $(q-1)\times(q-1)$ matrix

$$A_r(x,y)=e_q(rxy), \qquad x,y\in G_q. \tag{91}$$

Theorem 6.3 (Exact product-phase projective norm)**.** *For every $r\neq 0$,*

$$A_rA_r^* = qI_{q-1}-J_{q-1}, \tag{92}$$

and

$$\boxed{\pi_\infty(A_r)=\frac{(q-2)\sqrt{q}+1}{q-1}.} \tag{93}$$

Proof. For $x,x'\in G_q$, additive orthogonality gives

$$\sum_{y\in G_q} e_q(r(x-x')y)=\begin{cases} q-1, & x=x',\\ -1, & x\neq x'.\end{cases} \tag{94}$$

This is (92). Hence A_r has singular value $\sqrt{q}$ with multiplicity $q-2$ and singular value 1 with multiplicity one. The nuclear lower bound (39) gives the right side of (93).

For equality, use the multiplicative characters of G_q as an orthonormal basis. Every normalized character vector has all coordinates of modulus $(q-1)^{-1/2}$. The matrix A_r maps each such vector to a scalar multiple of the conjugate-character vector; the scalar is a Gauss sum of modulus $\sqrt{q}$ for nonprincipal characters and modulus 1 for the principal character. Its singular-value decomposition therefore has flat left and right singular vectors. Rescaling each vector to supremum one makes the total projective mass exactly $\|A_r\|_*/(q-1)$. $\square$

6.4 Absolute additive-frequency reassembly is costly

For $a\in G_q$, let

$$K_a(x,y)=\mathbf{1}_{xy=a}. \tag{95}$$

Proposition 6.4 (A cost-one product kernel)**.** *One has*

$$\boxed{\pi_\infty(K_a)=1.} \tag{96}$$

But its additive orthogonality expansion

$$K_a(x,y)=\frac{1}{q}\sum_{r\in\mathbb{F}_q} e_q(-ra)A_r(x,y) \tag{97}$$

has termwise absolute projective cost

$$\frac{1}{q}\left[1+(q-1)\frac{(q-2)\sqrt{q}+1}{q-1}\right] \asymp \sqrt{q}. \tag{98}$$

Proof. Multiplicative orthogonality gives

$$K_a(x,y) = \frac{1}{q-1}\sum_{\chi} \overline{\chi(a)}\chi(x)\chi(y), \tag{99}$$

whose projective mass is one. Since K_a has an entry equal to one, (36) gives equality. Formula (97) is ordinary additive orthogonality; insert theorem 6.3, noting that $A_0 = J_{q-1}$ has cost one. □

Corollary 6.5 (Cost of absolute additive-frequency separation)**.** *At $q \asymp J$, separating each nonzero additive frequency of $e_q(r\ell j)$ and then reassembling absolutely introduces a sharp amplitude cost $\asymp J^{1/2}$, hence an energy cost $\asymp J$. Thus this additive-frequency expansion cannot be bounded at the critical scale by summing its nonzero-frequency costs separately. The static mask may still be separated by theorem 4.4, while the multiplicative-character reconstruction* (99) *avoids the loss.*

This is a method-specific order-of-operations obstruction. It is not a lower bound for the physical orbit energy: the cost-one identity (99) demonstrates the cancellation that termwise additive separation discards.

7 Exact spectra for coherent complementary-factor sums

The fixed-column theorem reaches the product-slope shape, but the topology of the physical orbit slice is not the square function in s treated in TPC-35. We now diagonalize the operation that sums in s first.

7.1 The complete product-slope operator

Let $q > 2$ be prime, put $G_q = \mathbb{F}_q^{\times}$, and fix $h \in G_q$. Let $f, g : \mathbb{F}_q \to \mathbb{C}$ be additively centered:

$$\sum_{D\in\mathbb{F}_q} f(D) = 0, \qquad \sum_{U\in\mathbb{F}_q} g(U) = 0. \tag{100}$$

Define

$$R_s(a) := \sum_{D\in\mathbb{F}_q} f(D)g\Big(s^{-1}(aD+h)\Big), \qquad a, s \in G_q, \tag{101}$$

$$(Tb)(j,s) := \sum_{\ell\in G_q} b(\ell)R_s(\ell j), \qquad j, s \in G_q. \tag{102}$$

For a multiplicative character χ of G_q, use

$$\widehat{b}(\chi) = \sum_{x\in G_q} b(x)\overline{\chi(x)}, \qquad \widehat{R}_s(\chi) = \sum_{a\in G_q} R_s(a)\overline{\chi(a)}. \tag{103}$$

TPC-35 proves the square-function identity

$$\sum_{j,s} |(Tb)(j,s)|^2 = \frac{1}{q-1}\sum_{\chi} |\widehat{b}(\bar{\chi})|^2 \sum_{s} |\widehat{R}_s(\chi)|^2. \tag{104}$$

7.2 A weighted coherent spectrum

For an arbitrary weight $w : G_q \to \mathbb{C}$, put

$$S_w(j) = \sum_{s \in G_q} w(s)(Tb)(j, s). \tag{105}$$

Theorem 7.1 (Weighted coherent product-slope spectrum)**.** *With the preceding conventions,*

$$\boxed{\sum_{j \in G_q} |S_w(j)|^2 = \frac{1}{q-1} \sum_{\chi} |\widehat{b}(\bar{\chi})|^2 \left| \sum_{s \in G_q} w(s) \widehat{R}_s(\chi) \right|^2.} \tag{106}$$

Proof. For fixed s, the substitution $a = \ell j$ gives

$$\sum_{j \in G_q} (Tb)(j, s)\overline{\chi(j)} = \widehat{b}(\bar{\chi})\widehat{R}_s(\chi). \tag{107}$$

Multiply by $w(s)$, sum over s, and apply multiplicative Parseval in j. □

The square function (104) contains $\sum_s |\widehat{R}_s(\chi)|^2$. The coherent topology contains instead

$$\left| \sum_{s} w(s) \widehat{R}_s(\chi) \right|^2. \tag{108}$$

This is the exact character-level quantity that a physical theorem must control after static reassembly.

7.3 The complete complementary-factor trace

The complete weight $w = 1$ has an unexpectedly simple closed form.

Theorem 7.2 (Residue-zero boundary trace)**.** *For every $a \in G_q$,*

$$\boxed{\sum_{s \in G_q} R_s(a) = qg(0)f(-h/a).} \tag{109}$$

Consequently

$$\sum_{j \in G_q} \left| \sum_{s \in G_q} (Tb)(j, s) \right|^2 = \frac{q^2|g(0)|^2}{q-1} \sum_{\chi} |\widehat{b}(\bar{\chi})|^2 \left| \sum_{D \neq 0} f(D)\chi(D) \right|^2. \tag{110}$$

Proof. Fix a and let $D_0 = -h/a$. For $D = D_0$, the argument of g is zero for every s, so its contribution to the s-sum is $(q-1)f(D_0)g(0)$. For $D \neq D_0$, the element $aD + h$ is nonzero and $s^{-1}(aD + h)$ runs once through G_q. Additive centering gives

$$\sum_{U \in G_q} g(U) = -g(0). \tag{111}$$

Also $\sum_{D \neq D_0} f(D) = -f(D_0)$. The remaining contribution is therefore $f(D_0)g(0)$, proving (109).

Now

$$\sum_{s} (Tb)(j, s) = qg(0) \sum_{\ell \in G_q} b(\ell) f\left(-\frac{h}{\ell j}\right). \tag{112}$$

Taking the multiplicative transform in j, the substitution $D = -h/(\ell j)$ gives, up to the unit phase $\overline{\chi(-h)}$,

$$\widehat{b}(\bar{\chi}) \sum_{D \neq 0} f(D)\chi(D). \tag{113}$$

Parseval proves (110). □

Corollary 7.3 (Exact complete cancellation when $g(0) = 0$)**.** *If $g(0) = 0$, then*

$$\sum_{s \in G_q} (Tb)(j, s) = 0 \tag{114}$$

for every b and every j.

Ordinary additive centering does not force $g(0) = 0$. If an uncentered physical g is supported away from zero and is then centered by subtracting its mean, its new value at zero is generally the negative of that mean. The boundary trace must therefore be extracted, cancelled, or estimated; it cannot be deleted by notation.

7.4 Sharpness of complementary-factor Cauchy

The exact trace does not imply a uniform gain for all centered data.

Theorem 7.4 (Sharp coherent-s saturation)**.** *For every prime $q > 2$, there are nonzero centered f, g and b, with $f(0) = 0$, such that*

$$\boxed{\sum_j \left| \sum_s (Tb)(j, s) \right|^2 = (q - 1) \sum_{j,s} |(Tb)(j, s)|^2.} \tag{115}$$

Thus the Cauchy factor $\#G_q = q - 1$ is sharp even after the principal energy arising from $f(0)$ has been removed.

Proof. Choose a nonprincipal multiplicative character χ and extend it by zero. Put

$$f(D) = \overline{\chi(D)} \ (D \neq 0), \qquad f(0) = 0, \tag{116}$$

$$g(U) = \mathbf{1}_{\{0\}}(U) - \frac{1}{q}, \qquad b(\ell) = \overline{\chi(\ell)}. \tag{117}$$

Both f and g are additively centered. The constant part of g contributes zero to $R_s(a)$ by character orthogonality. The remaining delta selects $D = -h/a$, so

$$R_s(a) = \overline{\chi(-h)}\chi(a), \tag{118}$$

independently of s. Hence

$$(Tb)(j, s) = (q - 1)\overline{\chi(-h)}\chi(j), \tag{119}$$

also independently of s. Equality in Cauchy follows pointwise in j, and summing proves (115). □

Remark 7.5 (Two independent critical J-losses)*.* At $q \asymp J$, termwise additive separation of the product phase loses J in energy by corollary 6.5. Uniform Cauchy in the complete s-variable can also lose J by theorem 7.4. The mechanisms are distinct. The first is avoided by retaining product slopes; the second requires a bound for the actual weighted trace (108).

8 Endpoint ledger and the relocated coefficient gate

The static multiplier question is now resolved, but the physical energy still requires integer-to-finite-field completion in the correct coherent topology. This section records exactly what the new theorems save and what they do not.

8.1 Complementary dyadic localization and a large-modulus realization

By (21), there are fixed constants $0 < c_{\mathscr{D}} < C_{\mathscr{D}}$ such that

$$c_{\mathscr{D}} X \le N_{\ell d}(j) := \ell dj + h \le C_{\mathscr{D}} X \tag{120}$$

throughout the packet. Partition $T < u \le U_0$ into disjoint half-open dyadic blocks

$$\mathscr{U} := \{2^k T : k \ge 0,\ 2^k T < U_0\}, \qquad \mathcal{U}_U := (U, 2U] \cap (T, U_0], \qquad |\mathscr{U}| = O(\log X), \tag{121}$$

and attach to each block the complementary interval

$$\mathcal{S}_U := \left\{ s \in \mathbb{N} : \frac{c_{\mathscr{D}} X}{2U} \le s \le \frac{C_{\mathscr{D}} X}{U} \right\}. \tag{122}$$

On an equality solution $su = N_{\ell d}(j)$ with $u \in \mathcal{U}_U$, the indicator of $\mathcal{S}_U$ is identically one. Hence the exact kernel identity

$$\mathbf{1}_{T<u\le U_0} \mathbf{1}_{su=N_{\ell d}(j)} = \sum_{U \in \mathscr{U}} \mathbf{1}_{u \in \mathcal{U}_U} \mathbf{1}_{s \in \mathcal{S}_U} \mathbf{1}_{su=N_{\ell d}(j)} \tag{123}$$

holds pointwise. The new cutoffs are unary in u and s, and summing the $O(\log X)$ blocks costs only $X^{o(1)}$.

Now fix one such block and put

$$F = su - N_{\ell d}(j) = su - \ell jd - h. \tag{124}$$

On the full Cartesian support $u \in \mathcal{U}_U$, $s \in \mathcal{S}_U$, one has

$$su \le 2C_{\mathscr{D}} X, \qquad |F| \le 3C_{\mathscr{D}} X. \tag{125}$$

Enlarge a fixed constant C_0 so that $C_0 \ge 3C_{\mathscr{D}}$ and every individual physical variable is smaller than $C_0 X$. By Bertrand's postulate, for large X one may choose a prime

$$2C_0 X < q < 4C_0 X, \tag{126}$$

and then

$$F \equiv 0 \pmod q \quad \Longleftrightarrow \quad F = 0. \tag{127}$$

Thus, block by block, the modular product-slope layer is pointwise identical to the integer layer after the physical variables are zero-extended to $\mathbb{F}_q$. Summing the dyadic blocks preserves this exact identity at soft cost.

This is an unconditional algebraic realization, not an efficient analytic one. In the standard completion route, the J-long orbit occupies only a proportion

$$\frac{J}{q} \asymp \frac{J}{X} \asymp \frac{1}{Q} \tag{128}$$

of the large modulus. Any argument that forgets this localization before using a complete normalized estimate may pay the coverage scale Q. This is method-specific bookkeeping, not a universal lower bound for every large-modulus method.

8.2 Orbit-scale moduli and aliases

On each reciprocal dyadic block, at the useful scale $q \asymp J$, the congruence $F \equiv 0 \pmod q$ admits

$$O(1 + X/q) = O(Q) \tag{129}$$

integer aliases. The $O(\log X)$ blocks are absorbed into $X^{o(1)}$. TPC-35 proves the exact joint ladder: for pairwise coprime $q_i \asymp J$, $1 \le i \le r$, the alias count is

$$O\Big(1 + \frac{X}{J^r}\Big). \tag{130}$$

In particular,

$$\frac{X}{J^3} = \frac{Q}{J^2} = X^{1/400+o(1)}, \tag{131}$$

while four moduli give strict no-wrap. Equality (131) is only an exponent-compatible ledger. Reassembling every three-modulus alias at the diagonal scale and controlling all proper-coordinate character faces remain unproved.

On every block, the complementary factor has length at most

$$S_0 \ll X/T. \tag{132}$$

Reducing a bare interval of this length modulo $q \asymp J$ gives maximum residue multiplicity

$$\ll 1 + \frac{S_0}{q} \ll 1 + \frac{Q}{T} = X^{563/2000+o(1)}. \tag{133}$$

The actual weight is more structured than a bare interval, but (133) shows that completeness in s cannot be treated as cost-free.

8.3 Resolved and unresolved scales

Table 1: Status of the post-TPC-35 reassembly routes.

Object or operation	Scale or cost	Status
Actual static row mask	$X^{o(1)}$ projective	proved here
Fixed-column source–d separation	$O(D)$ layers, $X^{o(1)}$ mass	proved here
Signed Ferrers separation	$\Theta(\log D)$	proved sharp here
Nonnegative Ferrers separation	D	proved sharp here; avoid
Termwise additive product-phase split	$q^{1/2}$ amplitude	proved sharp for that route; avoid
Uniform complete-s Cauchy	q energy	proved sharp abstractly
Large-modulus standard coverage	$q/J \asymp Q$	valid but inefficient route
One orbit-scale modulus	$X/q \asymp Q$ aliases	deterministic envelope; too many
Three orbit-scale moduli	$X/J^3 = Q/J^2$	critical ledger; reassembly open
Actual weighted coherent spectrum	(108)	exact object; estimate open

Two formerly conflated questions are now separate. The first asks whether the mask can be put into signed product-ready form; the answer is yes. The second asks whether the resulting actual coefficient direction has small weighted coherent character mass after aliases and incomplete ranges are restored; that is the new gate.

8.4 A coefficient-specific coherent target

For one completed finite-field layer, define

$$\mathcal{G}_q(b, f, g; w) := \frac{1}{q-1} \sum_{\chi} |\widehat{b}(\bar{\chi})|^2 \left| \sum_{s} w(s) \widehat{R}_s(\chi) \right|^2 . \tag{134}$$

By theorem 7.1, this is exactly $\|S_w\|_2^2$, not a majorant. The next physical theorem can therefore be pursued through either of two nonexclusive routes:

(a) prove an aggregate bound for (134) over the actual reassembly layers and aliases; or

(b) prove a quantitative avoidance statement showing that the actual b, f, g, w have negligible overlap with the character pairs on which (108) is large.

Neither route asks for a maximum over all abstract coefficients. Conversely, theorem 7.4 shows that centeredness and $f(0) = 0$ alone cannot prove the desired aggregate estimate.

8.5 Route menu for the next layer

The following routes are nonexclusive.

Route A: residue-zero boundary extraction.
Split the complete s-trace (109) from the incomplete boundary weight. Determine whether the actual centered ultra coefficient has a small or diagonally absorbable $g(0)$ component before estimating the remainder.

Route B: weighted two-character dispersion.
Expand the physical s-weight multiplicatively and estimate $\sum_s w(s)\widehat{R}_s(\chi)$ by the actual Möbius character support, rather than by Cauchy. This is the closest continuation of the spectral route.

Route C: coherent three-modulus CRT.
Keep three orbit-scale moduli separate, center each coordinate face, and prove diagonal-scale reassembly simultaneously for principal, singleton, double, and full-coordinate modes. Static mask complexity no longer adds a polynomial loss.

Route D: a scoped negative theorem.
If the physical coefficient family can be shown to concentrate on the saturating s-coherent mode, or if a proper-coordinate alias survives every admissible centering, that is a precise obstruction to this route. It would not be a lower bound for twin-prime correlations themselves.

The first two routes now have exact finite-model targets; the third has the critical exponent budget. Failure of any one route leaves the others available without changing the results proved in this paper.

8.6 Relation to the parity boundary

The factor $q \asymp J$ appears twice: as the energy cost of splitting a product phase additively, and as the sharp uniform Cauchy cost in the coherent s-variable. Avoiding the first is an algebraic order-of-operations theorem. Avoiding the second requires arithmetic information about the actual coefficient family. No estimate here supplies that information.

Accordingly, the soft mask theorem is a genuine advance in the proof program, but it is not a parity-breaking theorem in the sense of classical sieve theory [1, 2]. It removes one candidate obstruction and exposes the next one in an exact norm.

9 Exact certificate and claim boundary

The accompanying standard-library certificate is

`experiments/tpc36_certificate.py`.

It uses integers, `Fraction`, and symbolic finite-group orthogonality. It contains no floating-point calculation, random input, true division, or Python `assert` statement. Running

```
python experiments/tpc36_certificate.py
python -O experiments/tpc36_certificate.py
```

performs the same checks and regenerates byte-identical JSON.

9.1 Finite exact checks

The archived run performs 137,083 checks.

Table 2: Exact check families in the TPC-36 certificate.

Check family	Count
Gcd projector and orthogonal compression	120,116
Source equality and inequality compression	970
Ferrers Fourier reconstruction and far-row endpoints	1,671
Triangular Gram, inverse, and eigen-polynomial identities	42
Product-phase Gram $qI - J$	3,350
Multiplicative permutation-kernel reconstruction	7,286
Coherent s-trace and sharpness example	3,644
Source constraints	4
Total	137,083

For the gcd suite, the script enumerates squarefree packets and cutoff values, verifies (55), and then verifies the Walsh analogue of (58) coefficient by coefficient. The source suite checks the normalized orthogonality identity and its complement. The Ferrers suite represents roots of unity symbolically, reconstructs every prefix entry, and checks (53) at both exact endpoints $m_\gamma - \Delta$ and $m_\gamma + \Delta$.

The triangular suite constructs $T_n T_n^*$, verifies its exact first-difference inverse, and compares the inverse characteristic polynomial with its recurrence for $n \le 7$. The product-phase suite counts exponent residues to verify (92); the multiplication suite uses discrete logarithms to check (99). The last suite enumerates centered finite-field data for (109) and checks the nonprincipal construction (116)–(119).

9.2 Reproducibility values

Line endings are normalized before the source digest is taken. The archived values are

normalized source SHA-256 `ed669550fbf29aecdc31062f98fc64463`

$$\texttt{67871505f46d6a4bf0e20d7625a7151}, \tag{135}$$

$$\text{JSON SHA-256} \quad \begin{array}{l} \texttt{9fbbc133d77ca53ed978e75461900917b} \\ \texttt{188bfdb9abf3bd436c1df5e4cfd09b6}, \end{array} \tag{136}$$

$$\text{certificate digest} \quad \begin{array}{l} \texttt{4718f7216d8e67a87e84e3ebf6c550af} \\ \texttt{ca725c27c7980070f11ca6025ce7ff85}. \end{array} \tag{137}$$

The certificate digest hashes the canonical payload before its digest field is appended.

9.3 What is proved

The unconditional new results of this paper are:

(i) the exact global orthogonal compression (58) and soft gcd projective bound;

(ii) the global actual-mask theorem (63);

(iii) the fixed-column source–divisor representation (72) with logarithmic signed mass;

(iv) the exact Ferrers singular values, the optimal signed logarithmic cost, and the exact nonnegative cost n;

(v) the product-phase Gram and exact projective norm (93);

(vi) the cost-one multiplication kernel and the sharp loss for its termwise additive-frequency reassembly;

(vii) the weighted coherent spectrum (106); and

(viii) the complete s-trace, its residue-zero energy formula, and the sharp centered nonprincipal Cauchy example.

The physical product-ready column theorem also uses the fixed residue-cell split and Mellin separation already proved in TPC-25. Its dependence on that prior result is explicit rather than re-proved here.

9.4 What is not proved

Finite enumeration certifies algebra on the tested families; it does not prove asymptotic analytic estimates. Neither the manuscript nor the certificate proves:

(i) an operator or Hilbert–Schmidt bound for the actual orbit-Gram off-diagonal;

(ii) the coefficient-specific physical gate $V_L + V_R \ll X^{o(1)}Q^3/J$;

(iii) a complete integer-to-modular reassembly at $q \asymp J$;

(iv) control of one-modulus or three-modulus alias fibers at diagonal scale;

(v) a bound for the actual weighted coherent functional (134);

(vi) vanishing of the actual centered residue-zero boundary $g(0)$;

(vii) cancellation of the terminal/proper-divisor cross terms;

(viii) an affine Chowla estimate or Möbius disjointness theorem;

(ix) a breach of sieve parity, positivity, or a Hardy–Littlewood prime-pair asymptotic; or

(x) infinitely many twin primes.

The precise advance is narrower and reusable: static physical-mask complexity is no longer an unresolved polynomial obstruction. Signed reassembly places the real next gate at weighted coherent spectral overlap and joint alias control. A positive or negative theorem about that new functional would be the next logically meaningful layer.

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